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Optimizing of the underground power cable bedding using momentum-type particle swarm optimization method

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ABSTRACT

Thermal performance optimization of underground power cable system is presented in this paper. The analyzed system consists of three underground power cables situated in an in-line arrangement. The HDPE (High-Density Polyethylene) casing pipes, filled with SBM (Sand-Bentonite Mixture), covers the cables to protect them from heavy mechanical loads (e.g. vibrations). The FTB (Fluidized Thermal Backfill) layer is applied to prevent the cables from overheating. Due to the substantial costs of FTB backfill material (in relation to the native soil or dry sand), the cross-sectional area of FTB bedding layer has to be minimized. Furthermore, the maximum cable conductor temperature is expected not to exceed the optimum operating temperature. Therefore, the optimization procedure i.e. momentum-type PSO (Particle Swarm Optimization) is applied. The FEM (Finite Element Method) is used to solve the two-dimensional steady-state heat conduction problem. As a result, temperature distribution is determined for the native soil, FTB bedding, and cables. The performed computations considered the temperature dependent current rating and volumetric heat generation rate from cable conductor. The applied optimization procedure resulted in determination of the optimum cable spacing and cross-sectional area of the rectangular-shaped FTB bedding layer. Moreover, the obtained maximum temperature for the cable core do not exceed the allowable value.

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1. Introduction

In recent years, the thermal analysis of underground power cable system attracted a broad scientific attention. In most instances, the electric energy transmission line operates at the maximum possible conductor electric current. According to first Joule's law, heat generated in power cable core depends on its electric resistance and flowing electric current. An accurate analysis of heat dissipation process from the underground power cables to the surrounding soil plays a crucial role in designing the electricity transmission lines in modern power plants [1,2].

Ampacity is defined as the maximum electrical current that a conductor can safely carry without exceeding its insulation temperature limitations. Power cable ampacity is also described as a current carrying capacity. The current carrying capacity mostly depends on the temperature of the cable conductor. Excessive conductor temperature leads to cable overheating and the improper operation of the power transmission line. Furthermore, the better conditions of heat dissipation process from a cable to its surroundings, the lower cable diameter may be used for the same electrical load. Therefore, the unit costs associated with underground transmission line installation decreases significantly.

The traditional method used in calculations of the thermal resistance between the cable system and the external environment [3,4] assumes that the soil is a homogeneous material with constant thermal conductivity. In fact, the heat transfer processes associated with heat dissipation from the power cable to the surrounding soil are more complex. The soil is a multilayered porous material and consists of e.g. quartz, organic matter, clay minerals, air, water in the liquid and vapor phase, among others. Moreover, the heat transfer conditions depend strongly on the thermal conductivity of each soil layer. Thus, soil thermal conductivity exhibits a distinct dependence on porosity, liquid vapor transport, and temperature, among others [5]. The thermal conductivity increases with soil porosity since the pores are filled with water and the air

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Nomenclature

A	cross-section area (m^2)
b	distance between conductor axis and top edge of the FTB bedding layer (m)
d	diameter (m)
f	alternating current (AC) frequency (Hz)
$F(\mathbf{x})$	cost function
H	burial depth (m)
I	current load (A)
k	thermal conductivity (W/(m K))
l	distance between conductor axes (m)
p	distance between conductor axis and bottom edge of the FTB bedding layer (m)
q_v	heat source per unit volume (W/m^3)
rand()	random number between 0 and 1 (−)
R_e	electrical resistance (Ω/km)
R_{th}	thermal resistance ((m K)/W)
r	radius (m)
s	distance between conductor axis and lateral edge of the FTB bedding layer (m)
T	temperature (°C)
T_{cmax}	maximum central cable core temperature (°C)
$T_{c,FEM}$	cable core temperature determined from FEM model (°C)
$T_{max,p}$	maximum allowable cable core temperature (°C)
$\mathbf{v}_i^n$	i-th particle velocity in n-th iteration (m/s)
$\mathbf{x}$	design variables vector (or particle position vector) (m)
$\mathbf{x}_i^n$	i-th particle position in n-th iteration (m)
y_s	skin effect factor (−)
y_p	proximity effect factor (−)

Greek symbols

α	temperature coefficient (−)
ΔQ	heat losses from the power cable per unit length (W/m)

Subscripts

AC	alternating current
b	cable bedding layer
c	cable conductor
DC	direct current
g	at ground level
ins	XLPE cable insulation
ja	cable jacket
max	maximum value (related to the central cable core temperature T_{cmax} or cost function $F(\mathbf{x})$)
mean	mean value (related to the central cable core temperature T_{cmax} or cost function $F(\mathbf{x})$)
min	minimum value (related to the central cable core temperature T_{cmax} or cost function $F(\mathbf{x})$)
opt	optimum cable core temperature
pf	constants of dynamic penalty function $C(n)$ and the penalty factor $H(\mathbf{x})$
PSO	constants of particle swarm optimization (PSO) algorithm
ref	reference value
sh	cable sheath
soil	soil

Superscripts

n	iteration number

List of shorts

FTB	fluidized thermal backfill
FEM	finite element method
PSO	particle swarm optimization
HDPE	high-density polyethylene
SBM	sand-bentonite mixture
PE	polyethylene
PVC	polyvinyl chloride
SGFC	sand, gravel, fly ash and cement-mix
IEC	international electrotechnical commission
XLPE	cross-linked polyethylene

[6,7]. The higher soil water content, the better heat dissipation process conditions occur. However, water existing in pores may locally evaporate in the immediate vicinity of the cable thus the so-called 'dry zones' are created. Dry zones are usually formed around underground power cables under loading conditions due to the moisture migration within the soil [8]. The presence of dry zones around the cable results in a significant soil thermal conductivity drop since the thermal conductivity is over 20 times lower for a vapor than for liquid. Therefore, when temperature is rising, and dry zones are formed, it leads to a significant reduction in soil thermal conductivity.

Geometrical configuration of the cable system (i.e. cable spacing, shape of the trench, backfill layer cross-sectional area) influences the cable core temperature. In thermal engineering practice, it is possible to enhance the heat transfer from the solid materials to its surroundings by different methods. In particular cases: extended surfaces [9–11], porous structures [12–14] and multiphase materials [15,16] are utilized. During the underground transmission line placement, the heat dissipation from power cables to the surrounding soil shall be enhanced.

In order to circumvent the power cable overheating problems the specially designed bedding material, e.g. FTB (Fluidized Thermal Backfill), exhibiting higher thermal conductivity than the mother ground, is often used. Moreover, the thermal conductivity of the dry FTB material is over three times greater than for the dry soil. Therefore, underground power cables placement in the FTB bedding instead of the mother ground may circumvent the problems associated with a considerable thermal conductivity drop in soil, especially in the vicinity of the cable. On the other hand, the material costs related to FTB backfilling are significantly higher, comparing to the mother ground. Approximate cost of FTB is about $130 per cubic meter [17] and strongly depends on the cable line installation location and availability of raw materials, among others. Hence, when designing the shape of FTB layer (the optimum cross-sectional area of the bedding layer), the optimization work has to be done.

This paper presents an optimization of the FTB bedding layer cross-sectional area for underground power cable system that is planned to be installed in one of the Polish power plants. The analyzed system consists of three underground power cables situated in an in-line arrangement. Minimization of the FTB bedding cross-sectional area is performed by employing the momentum-type PSO (Particle Swarm Optimization) Method [18]. Moreover, the cable conductor temperature cannot exceed the optimum operating temperature i.e. 65 °C, which is specified by the cable producer. The Finite Element Method [5,19,20,23,24] is applied to determine the temperature distribution in underground power cable system. The methods for thermal analysis of underground power cable systems can be also found in Refs. [21,22].

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2. Solutions for installation of underground cable systems

Direct burial power cable laying technique is widely used in most countries due to the low operational costs and immunity to adverse weather conditions. Application of light mortar or thermal filler instead of fine sand may considerably improve the circuit transmission properties. The power cable transmission line may be laid in the ground in a different configurations [25]. Typical formations include trefoil (triangular) and flat (in-line) arrangement (Fig. 1). The appropriate type of cables arrangement depends on several factors like screen bonding method, conductor area and available space for installation.

FTB is one of the commonly used and popular backfill materials. It is the engineered slurry backfill mixture which, when solidified, may develop into an efficient heat conducting medium [26].

The underground power cables are commonly situated in ducts. This laying method has a significant advantage over the conventional burial technique. The power cables are usually located in the PVC (Polyvinyl Chloride) or PE (Polyethylene) casing pipes. Cable placement in ducts is widely applied when the ground is subjected to unusually heavy loads (i.e. significant vibrations) e.g. under crossings level (Fig. 1). Flat formation is applied to particular cases i.e. protected cables: 225 and 400 kV auxiliary, under road crossings [27]. The PVC or PE casing pipes are filled with Sand-Bentonite Mixture (SBM). SBM improves the heat dissipation from hot cable core to the surrounding soil by increasing the thermal conductivity in the vicinity of the cable line [28].

Bentonite, which is commercially available in powder form, is a swelling montmorillonite clay which, when mixed with water, expands and forms a gel. In gel form, it acts as a lubricant or fluidizer for the mixture [29]. Thermal conductivity of SMB mainly depends on three parameters i.e. mixture constituents (sand/bentonite) proportions, dry density, and water content [30]. An increase in the SBM water content and dry density improves thermal conductivity of the mixture [31]. Also, the proper proportion of individual components influence the SBM thermal properties. Due to the swelling of bentonite particles the better contact between the grains is achieved.

This paper presents the thermal analysis of 400 kV high voltage underground power cables [32] placed in the flat formation. Table 1 gives all the power cable relevant parameters. According to the producer requirements [33] the maximum allowable cable core temperature shall not be exceeded. When the cable core temperature is greater than 90 °C, the XLPE cable insulation melting may occur leading directly to the transmission line malfunction. The optimum power cable operating temperature is specified as 65 °C.

Fig. 2 depicts the structure of considered high voltage power cable with a conductor diameter of 49.6 mm and cross-sectional area of 1600 mm^2.

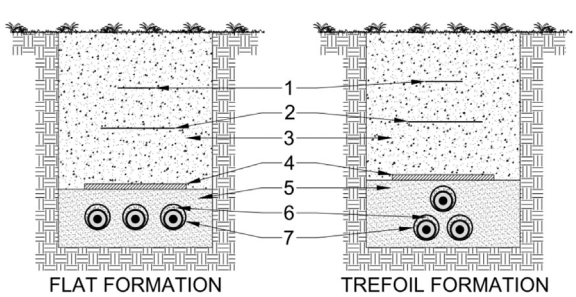

Fig. 1. Types of buried cables installation in ducts; 1 – warning tape, 2 – warning grid, 3 – mother ground, 4 – concrete protective slab, 5 – FTB, 6 – SBM, 7 – PVC or PE casing pipe.

Table 1
400 kV power cable characteristics [33].

Cross-section area, mm^2	1600
Conductor diameter d_c, mm	49.6
Insulation total thickness, mm	27
External diameter, mm	127.9
DC Resistance at 20 °C, Ω/km	0.0113
AC Resistance at 65°C/50 Hz, Ω/km	0.0157
Current loading in ground in-line at 65 °C, A	1145

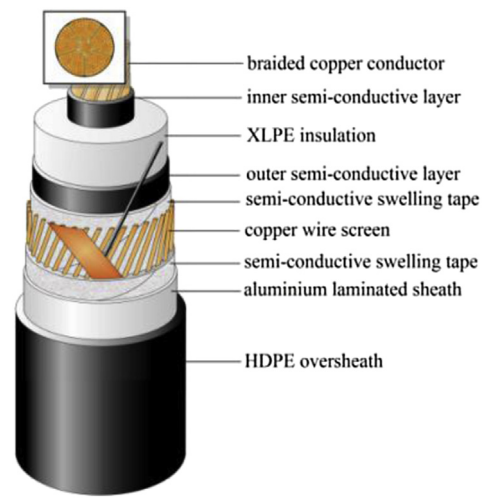

Fig. 2. 400 kV high-voltage power cable layout [33].

Table 2 lists the materials, thermal conductivities and thicknesses of the individual cable components.

3. Optimization of cable bedding layer cross-sectional area

The transmission line is considered to be installed beneath the road crossing. The optimal shape of the cable bedding trench is determined under the conditions of minimizing the FTB bedding layer cross-sectional area while the cable temperature shall not exceed 65 °C. The momentum-type Particle Swarm Optimization (PSO) method is used for this purpose.

Table 2
Thermal properties and thicknesses of 400 kV power cable layout materials (referred to Fig. 2).

Cable constituent	Material	Thickness, mm	Thermal conductivity, W/(m K)
Conductor	copper	⌀49.6	400.00
Inner semi-conductive layer	–	2.0	0.2875
Insulation	XLPE	27.0	0.2875
Outer semi-conductive layer	–	1.5	0.2875
Semi-conductive swelling tape as longitudinal water barrier	–	0.7	0.2875
Wire screen	copper	4.8	400.00
Semi-conductive swelling tape as longitudinal water barrier	–	0.7	0.2875
Aluminum laminated sheath	aluminium	0.2	0.2875
HDPE oversheath	HDPE	5.1	0.2875

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Table 3
Momentum-type PSO constants and initial values of variables, based on [18].

Parameter	Value
β_{PSO}	0.1
α_{PSO}	1
c_1	2
c_2	2
$\Delta\mathbf{v}_i^0$ (initial value of delta momentum term)	0

3.1. Momentum type Particle Swarm Optimization (PSO)

This study employs a momentum-type PSO algorithm [18] and a dynamic penalty function with a continues assignment to solve a discrete nonlinear constrained global optimization problem. As it has been shown in Refs. [18], the momentum-type PSO may achieve faster convergence than the traditional PSO algorithm [34–36]. The PSO is inspired by the social behavior simulations for groups of organism, such as flocks of birds and schools of fish. The particles represent the sets of design variables. The particle swarm with size N_{par} is defined during the initial iteration. The particle velocity determines its position in successive iteration. The algorithm finds the global minimum of the cost function$f(\mathbf{x})$ if the particle position is the same for all particles in the swarm, or the other specified convergence criterion is satisfied. The momentum type PSO algorithm uses only two equations, first for the velocity of i-th particle in n-th iteration $\mathbf{v}_i^n$ and the second for the new position of the particles $\mathbf{x}_i^n$.

The general form of the equation for i-th particle velocity $\mathbf{v}_i^n$ in n-th iteration is given as follows [18]:

$$\mathbf{v}_i^n = \beta_{PSO} \times \Delta\mathbf{v}_i^{n-1} + c_1 \times rand() \times \left(pbest_i - \mathbf{x}_i^{n-1}\right) + c_2 \times rand() \times \left(gbest_i - \mathbf{x}_i^{n-1}\right). \quad (1)$$

The particle position $\mathbf{x}_i^n$ is updated in the following manner:

$$\mathbf{x}_i^n = \mathbf{x}_i^{n-1} + \alpha_{PSO} \times \mathbf{v}_i^n \quad i = 1, 2, ..., N_{par}. \quad (2)$$

The momentum type PSO algorithm parameters are given in Table 3. The delta momentum term in the applied PSO algorithm $\Delta\mathbf{v}_i^{n-1} = \mathbf{v}_i^{n-1} - \mathbf{v}_i^{n-2}$ stabilizes searching, and an appropriate value for β_{PSO} improves the convergence and optimal solution [18].

In a particle velocity formula given by Eq. (1), $rand()$ is a random number between 0 and 1. In a state of equilibrium, other particles are not able to find better positions, i.e. $\mathbf{x}_i^n = pbest_i$ and $\mathbf{x}_i^n = gbest_i$ where $pbest$ denotes the best particle position, and $gbest$ is the global best position. The converged algorithm solution ensures that the Eq. (1) satisfies the zero velocity variation($\Delta\mathbf{v}_i^{n-1} = 0$) and zero velocity ($\mathbf{v}_i^n = 0$) conditions for the considered particle.

In order to handle the nonlinear constraints the dynamic penalty function, as suggested in Refs. [18], shall be applied. An active function $C(n)$ and a continuous assignment function$H(\mathbf{x})$ are multiplied and added to the original cost function $f(\mathbf{x})$:

$$F(\mathbf{x}) = f(\mathbf{x}) + C(n)H(\mathbf{x}), \quad (3)$$

where $C(n)$ denotes a dynamically modified penalty value dependent on the iteration number n; $H(\mathbf{x})$ is a penalty factor. The terms $C(n)$ and $H(\mathbf{x})$ for m linear and nonlinear constraint $g_j(\mathbf{x})$ are given as [18]:

$$C(n) = \left(c_{pf} \times n\right)^{\alpha_{pf}}, \quad (4)$$

$$H(\mathbf{x}) = \sum_{j=1}^{m} \left[\theta\left(q_j(\mathbf{x})\right) \times q_j(\mathbf{x})^{\gamma\left(q_j(\mathbf{x})\right)}\right], \quad (5)$$

$$\theta\left(q_j(\mathbf{x})\right) = a_{pf} \times \left(1 - \frac{1}{e^{q_j(\mathbf{x})}}\right) + b_{pf}, \quad (6)$$

where $q_j(\mathbf{x}) = \max\{0, q_j(\mathbf{x})\}, j = 1, ..., m$. The function $q_j(\mathbf{x})$ is a relative violated function of the constraints, $\theta(q_j(\mathbf{x}))$ denotes a continuous assignment function of exponential type, and $\gamma(q_j(\mathbf{x}))$ defines the power of the violated function. The constants for equation (4)÷(6) are selected as: $c_{pf} = 0.5$, $\alpha_{pf} = 2$, $a_{pf} = 150$, and $b_{pf} = 10$. If $q_j(\mathbf{x}) \leq 1$, then the power $\gamma(q_j(\mathbf{x})) = 1$; in the other case $\gamma(q_j(\mathbf{x})) = 2$, as given in Ref. [18].

3.2. Optimization problem formulation

This study applies the momentum-type PSO for minimizing the cross-sectional area A_b of FTB layer. It should be noted that A_b is, in fact, only a half of the real FTB cross-sectional area. This assumption is valid since the physical geometry of the modeled underground power cable system and the expected pattern of the thermal solution, exhibits mirror symmetry. Fig. 3a shows the design variables l, s, p and b, where l is the horizontal distance between the cable conductors axes, s is the spacing between the right edge of the FTB bedding layer and the side cable axis. The distances between the conductors axes and the top and the bottom of FTB layer are

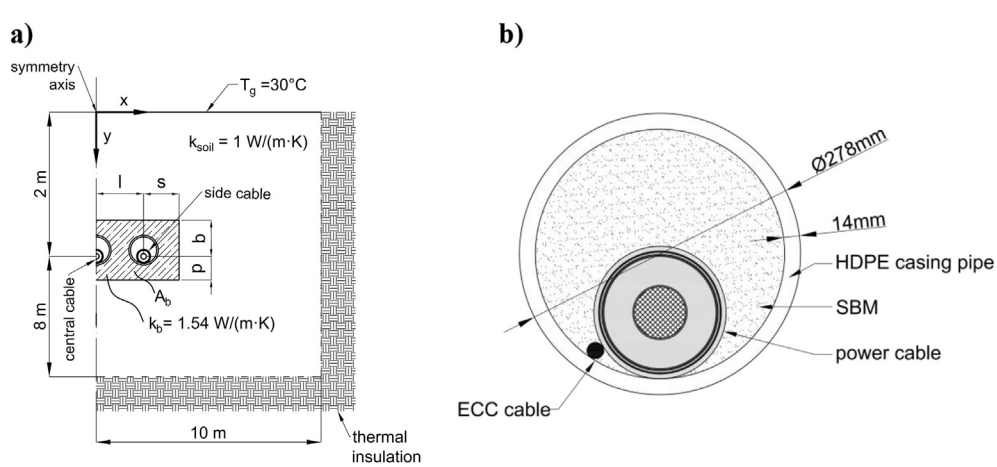

Fig. 3. Flat formation arrangement of a) underground power cable system, subjected to the optimization procedure; b) detailed view on cable placement in HDPE casing pipe.

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denoted as b and p, respectively. In order to avoid the excessive mechanical loads, the cables are situated in casing pipes made of High-Density Polyethylene (HDPE). The casing pipe (Fig. 3b) is filled with the Sand-Bentonite Mixture (SBM).

Equations (7) and (8) define the optimization problem for the underground cable system (Fig. 3):

Minimize

$$F(\mathbf{x}) = A_b(\mathbf{x}) + C(n)H(\mathbf{x}), \quad \mathbf{x} = [l,\ s,\ p,\ b] \in R^{N_{dim}=4} \tag{7}$$

Subject to

$$\begin{aligned} &q_j(\mathbf{x}) \le 0, \qquad j = 1 \quad , \\ &\text{with } q_j(\mathbf{x}) = T_{\text{cmax}}(\mathbf{x}) - T_{opt} \quad , \end{aligned} \tag{8}$$

where T_{opt} is the optimum operation temperature of the cable conductor i.e. $T_{opt} = 65\ ^{\circ}\text{C}$ and $T_{\text{cmax}}(\mathbf{x})$ denotes the maximum cable conductor temperature, determined for the $\mathbf{x}$ vector of design variables. It should be noted that the number of function constraints j is equal to 1, since only the constraint is that the $T_{\text{cmax}}(\mathbf{x})$ temperature shall not exceed the T_{opt} value. In FEM simulations (subsection 4.2) the discrete model of underground power cable system is created based on the values of design variables specified by a vector $\mathbf{x}$. The $T_{\text{cmax}}(\mathbf{x})$ temperature is determined at the location of central cable conductor gravity center (Fig. 3), from FEM simulations. The constraints on design variables are assumed as:

$$0.3 \le\ l\ \le 0.6, \qquad 0.2 \le\ p, b, s\ \le\ 0.4\ . \tag{9}$$

For the given number of particles in swarm N_{par}, the $T_{\text{cmax}}(\mathbf{x}_i)$ temperatures and the corresponding FTB bedding layer cross-sectional areas $A_b(\mathbf{x}_i)$ are determined for each particle position $\mathbf{x}_i$; $i = 1,..., N_{par}$. As a result, the objective function $F(\mathbf{x}_i)$ is evaluated for all particles. Then the momentum PSO optimization procedure is applied to solve the constrained optimization problem given by Eqs. (7)–(9). The detailed application of momentum PSO algorithm for the analyzed underground power cables system is discussed in subsection 5.

4. Determination of the temperature distribution in underground power cable system

The FEM (Finite Element Method) is used to determine the temperature distribution in the thermal system presented in Fig. 3. The analyzed power cable system consists of three 400 kV single core underground power cables situated in the HDPE casing pipes, filled with SBM. The flat formation is considered. The casing pipes are placed in FTB layer with rectangular-shaped cross-section, in order to prevent the power cables from overheating. It is assumed that the FTB layer is buried in the mother ground with constant thermal properties. As above-mentioned, since the geometrical symmetry exists thus, only a half of the real thermal system (Fig. 1) is considered.

4.1. Material properties

The mother ground thermal conductivity k_{soil} is assumed to be constant and equal to 1.00 W/(m K), according to IEC standards [3]. In this case application of SGFC mixture (Sand, Gravel, Fly ash and Cement-mix), is proposed. SGFC is one of the FTB types with a given composition: 41% of fine aggregate, 49% of coarse aggregate, 2.5% of cement and 7.5% of fly ash (fluidizer). When dry conditions occur, SGFC mixture is characterized by dry density of 2187 kg/m^3 and thermal conductivity of 1.54 W/(m K) [29].

Table 4
Thermal conductivities of different materials considered in computations.

Material	Thermal conductivity, W/(m K)
Mother ground	IEC standards [1]: 1.00
FTB	dry conditions: 1.54
SBM	dry conditions: 0.95
HDPE casing pipe	0.48

The power cables are situated in HDPE casing pipes filled with sand-bentonite-based buffer material with a dry density of 1700 kg/m^3 and thermal conductivity of 0.95 W/(m K) [37]. The external diameter and thickness of HDPE casing pipe are 278 mm and 14 mm, respectively. Mixture proposed by Ref. [37] is composed of MX-80 (Na-bentonite), sand and water. Sand mass fraction and water contents are 37.5% and 8.17%, respectively [37]. Table 4 shows the thermal conductivity of different materials considered in this study.

The computational study considers the dry conditions of FTB and SBM backfills, what ensures the safety margin when the power cable core temperature is calculated. In real conditions, the heat dissipation process is more efficient, due to the moisture existing in the soil and backfill pores [38,39].

4.2. Computational procedure

The heat transfer problem (Fig. 3) is considered as two-dimensional and steady-state. The Finite Element Method code developed in MATLAB [41] is used to solve the heat conduction equation [6,43–45]:

$$\frac{\partial}{\partial x}\left[k\frac{\partial T}{\partial x}\right] + \frac{\partial}{\partial y}\left[k\frac{\partial T}{\partial y}\right] = -q_v. \tag{10}$$

In Equation (10), T denotes the temperature at any point in the x-y plane around the underground cable; k represents thermal conductivity, and q_v is a heat source per cable core unit volume. The heat source term q_v is evaluated as follows:

$$\begin{aligned} &q_v = q_v(T_{c,FEM}), && \text{for the cable conductor domain} \\ &q_v = 0, && \text{otherwise.} \end{aligned} \tag{11}$$

The $T_{c,FEM}$ denotes an averaged temperature within the finite element that belongs to the cable conductor domain, calculated as an arithmetic mean of nodal temperatures. It should be noted, that $T_{c,FEM}$ temperatures are determined for all finite elements that belong to the cable conductor domain of FEM model. In other domains of the FEM model i.e. cable insulation, cable sheath, cable jacket, FTB bedding layer and native soil no heat source exist, i.e. $q_v = 0$. In Eq. (11) $q_v(T_{c,FEM})$ is a temperature dependent volumetric heat source per cable core unit volume, defined as:

$$q_v(T_{c,FEM}) = \frac{\Delta Q}{A_c}, \tag{12}$$

where A_c denotes the cable conductor cross-section area. The value of A_c is calculated on the basis of FEM model of cable conductor as $A_c = 0.25\pi d_c^2$ while ΔQ denotes the cable heat losses (also called as Joule losses). Since the volumetric heat sources within finite elements, that represents the cable conductor domain i.e. $q_v(T_{c,FEM})$, depend on the corresponding $T_{c,FEM}$ temperatures, thus Eq. (10) is solved iteratively to determine $T_{c,FEM}$ temperatures. Then for the cable conductor domain the values of $T_{c,FEM}$ from the previous iteration are used when determining the values of $q_v(T_{c,FEM})$ in the actual iteration.

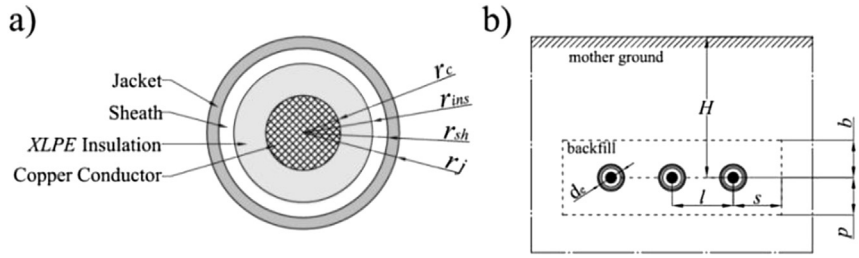

Fig. 4. Simplified model of a) 400 kV underground power cable considered whilst determination of volumetric heat transfer rate q_v; b) the geometrical arrangement of transmission line.

The cables are located at depth H of 2 m below the road crossing level. The square domain with height and width of 10 m is used in the computations (Fig. 3a). It is assumed that the right, left and bottom edges of the boundary region are thermally insulated (i.e. no heat flow exists in a direction perpendicular to the domain boundary). At the top edge (the ground level), the temperature T_g is set to 30 °C and is 5 °C higher than in Ref. [40] due to the possible adverse conditions of heat dissipation. Equation (13) summarizes the boundary conditions types applied when solving Eq. (10):

$$\begin{aligned}
&\left.\frac{\partial T}{\partial x}\right|_{x=10\mathrm{m}} = 0, && \text{for the boundary region right edge,}\\
&\left.\frac{\partial T}{\partial x}\right|_{x=0} = 0, && \text{for the symmetry plane,}\\
&\left.\frac{\partial T}{\partial y}\right|_{y=10\mathrm{m}} = 0, && \text{for the boundary region botom edge,}\\
&T(x, y=0) = T_g = 30^\circ\mathrm{C}, && \text{for the boundary region top edge.}
\end{aligned} \tag{13}$$

4.2.1. Determination of power cable heat losses

The evaluation of the power cable heat losses ΔQ is necessary to determine the volumetric heat source $q_v(T_{c,FEM})$, defined by Eq. (12) and used when solving steady-state heat conduction problem, given by Eqs. (10) and (13). In the present calculations, the simplified geometrical model of 400 kV underground power cable (Fig. 4a) is used.

The cable is buried at the depth of $H = 2$ m from the ground level (Fig. 4b). Symbols r_c, r_{ins}, r_{sh} and r_j (Table 5) denote the cable conductor, insulation, sheath and jacket outer radiuses, respectively. The cable external diameter is $d_e = d_j = 2r_j$, and the conductor diameter is $d_c = 2r_c$ (Fig. 4a).

The power cable heat losses are expressed by Joule's law introducing alternating current (AC) electric resistance:

$$\Delta Q = I^2 R_{e,AC}. \tag{14}$$

Cable conductor exhibits a greater resistance to the flow of alternating current $R_{e,AC}$ than it does to direct current $R_{e,DC}$. The three major factors for the observed increase are skin and proximity effects of closely spaced current carrying conductors, and also increase in cable conductor temperature [25]. The increase in electrical resistance is expressed as:

$$\begin{aligned}
R_{e,AC} &= R_{e,DC}\left(1 + y_s + y_p\right)\\
&= R_{e,ref}\left[1 + \alpha_{ref}\left(T_{c,FEM} - T_{ref}\right)\right]\left(1 + y_s + y_p\right),
\end{aligned} \tag{15}$$

where $R_{e,ref}$, in Ω/km, and $\alpha_{e,ref}$ are the reference cable conductor electrical resistance, and the temperature coefficient for the conductor material, both given at the reference temperature $T_{ref} = 20$ °C (according to the cable producer [33]). The cable conductor temperature, determined from Eq. (10) for FEM model, is denoted as $T_{c,FEM}$ and given in °C. For the described computational cases (braided copper conductor) $R_{e,ref} = 0.0113$ Ω/km and $\alpha_{e,ref} = 0.00393$. Skin and proximity effect factors y_s and y_p [25] are calculated, respectively as:

$$y_s = \frac{x_s^4}{192 + 0.8x_s^4}, \quad x_s^2 = \frac{8\pi f}{R_{e,DC}} 10^{-7} k_s, \tag{16}$$

$$\begin{aligned}
y_p &= \frac{x_p^4}{192 + 0.8x_p^4}\left(\frac{d_c}{l}\right)^2\left[0.312\left(\frac{d_c}{l}\right)^2 + \frac{1.18}{\frac{x_p^4}{192+0.8x_p^4} + 0.27}\right],\\
x_p^2 &= \frac{8\pi f}{R_{e,DC}} 10^{-7} k_p,
\end{aligned} \tag{17}$$

where k_s and k_p values equals to 0.435 and 0.37 [42]; f is altering current frequency equal to 50 Hz.

Maximum current loading of the analyzed underground power cable system is assumed to be equal to 1145 A. The values of q_v as a function of cable conductor temperature are shown in Fig. 5.

The computations, presented in this paper, do not include directly the dry zones formation effect in a vicinity of cable [8] that can cause a rapid increase in cable core temperature. Therefore, for a given temperature, the corresponding q_v values calculated from Eq. (12), are increased by 10% (Fig. 5). Hence, the safety margin is provided for a design calculation of the presented power cable system. The function $q_v(T_{c,FEM})$, considered in the present calculations, for cable spacing $l = 0.4$ m, is depicted in Fig. 5 and given by the formula:

$$q_v(T_{cFEM}) = 0.03062T_{cFEM} + 8.7836. \tag{18}$$

For different values of cable spacing l, varying from 0.3 m to 0.6 m, the volumetric heat source was calculated by using the same procedure.

Table 5
Thermal properties and thicknesses of 400 kV power cable layout materials [45].

Cable layout	Material	Thickness, mm	Layer radius, mm	Thermal conductivity W/(m K)
Conductor	copper	$d_c = 49.6$	$r_c = 24.8$	400
Insulation	XLPE	30.5	$r_{ins} = 55.3$	0.2875
Sheath	copper	6.4	$r_{sh} = 61.7$	400
Jacket	HDPE	5.1	$r_j = 66.8$	0.2875

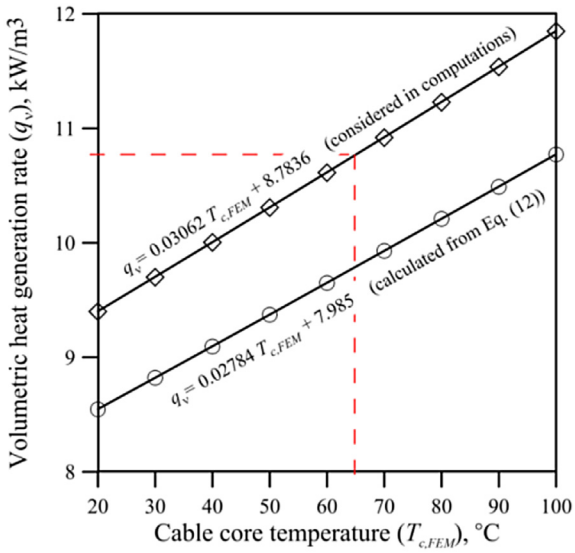

Fig. 5. Volumetric heat generation rate q_v as a function of the cable core temperature $T_{c,FEM}$ calculated for cable spacing $l = 0.4$ m.

The developed FEM code uses a temperature dependent $q_v(T_{c,FEM})$ tabularized data when solving the Eq. (10). Due to the temperature dependent heat generation rate the nonlinear system of equations is obtained. The Jacobi Iteration method [45] is applied to solve the nonlinear system and determine the nodal temperatures. The ground level temperature ($T_g = 30$ °C) is assumed as the starting values for Jacobi Iteration procedure. The discrete model of underground power cable system consists of triangular elements. The mesh is refined in the domains of FTB and SBM layer, HDPE casing pipes, and the cables, to follow the temperature gradients accurately.

5. Results and discussion

From the economic reasons, the FTB bedding layer cross-sectional area shall be minimized while ensuring the conductor temperature close to $T_{opt} = 65$ °C. Therefore, the momentum-type PSO algorithm, described in subsection 3.1, is applied to cable bedding area optimization problem given by Eqs. (7)–(9).

Presented study performs the momentum-type PSO for $N_{par} = 10$ and $N_{par} = 30$ number of particles. The design variables vector $\mathbf{x}_i^n = [p_i^n, b_i^n, s_i^n, l_i^n]$ defines the particle position during n-th iteration. The presence of side constraints in problem formulation allows a natural definition of scaled variables. The *upper* and *lower* bounds defined by Eq. (9) are used to scale the design variables vector $\mathbf{x}_i$ between 0 and 1:

$$\widehat{\mathbf{x}}_i^n = \left[\widehat{p}_i^n, \widehat{b}_i^n, \widehat{s}_i^n, \widehat{l}_i^n\right], \qquad \text{for } i = 1, \ldots, N_{par}, \tag{19}$$

where

$$\begin{aligned} \widehat{p}_i^n &= \frac{p_i^n - p_{lower}}{p_{upper} - p_{lower}}, \quad \widehat{b}_i^n = \frac{b_i^n - b_{lower}}{b_{upper} - b_{lower}}, \\ \widehat{s}_i^n &= \frac{s_i^n - s_{lower}}{s_{upper} - s_{lower}}, \quad \widehat{l}_i^n = \frac{l_i^n - l_{lower}}{l_{upper} - l_{lower}}. \end{aligned} \tag{20}$$

The rescaling of design variables vector $\widehat{\mathbf{x}}_i^n$ to $\mathbf{x}_i^n = [p_i^n, b_i^n, s_i^n, l_i^n]$ is performed as given below:

$$\begin{aligned} p_i^n &= \widehat{p}_i^n\left(p_{upper} - p_{lower}\right) + p_{lower}, \\ b_i^n &= \widehat{b}_i^n\left(b_{upper} - b_{lower}\right) + b_{lower}, \\ s_i^n &= \widehat{s}_i^n\left(s_{upper} - s_{lower}\right) + s_{lower}, \\ l_i^n &= \widehat{l}_i^n\left(l_{upper} - l_{lower}\right) + l_{lower}. \end{aligned} \tag{21}$$

In the developed computer code, the momentum-type PSO algorithm is implemented as follows. Firstly, the number of particles in a swarm, i.e. N_{par}, is defined. During the initial iteration ($n = 0$) the momentum-type PSO algorithm is initialized by performing the following steps:

1. Creation of random vectors of scaled design variables $\widehat{\mathbf{x}}_i^0$ for all particles existing in the swarm, i.e. $i = 1,\ldots,N_{par}$.
2. Setting the delta momentum term as $\Delta\mathbf{v}_i^0 = 0$, for $i = 1,\ldots,N_{par}$.
3. Rescaling of the random design variables vectors $\widehat{\mathbf{x}}_i^0$ to $\mathbf{x}_i^0$, for $i = 1,\ldots,N_{par}$, using Eq. (21).
4. Creation of the underground power systems FEM models based on the values of the rescaled design variables vectors $\mathbf{x}_i^0 = [p_i^0, b_i^0, s_i^0, l_i^0]$ for $i = 1,\ldots,N_{par}$.
5. Solving the steady state heat conduction problem, Eq. (10), for determining the maximum temperatures of central cable conductor $T_{\text{cmax}}(\mathbf{x}_i^0)$, for $i = 1,\ldots,N_{par}$.
6. Evaluation of the cost function $F(\mathbf{x}_i^0)$ for $i = 1,\ldots,N_{par}$ specified by Eq. (7), with constraints given by Eqs. (8) and (9).
7. Calculation of the maximum ($T_{max}^{n=0}$), mean ($T_{mean}^{n=0}$) and minimum ($T_{min}^{n=0}$) values of maximum central cable conductor $T_{\text{cmax}}(\mathbf{x}_i^{n=0})$ temperatures for all particles:

$$\begin{aligned} T_{max}^n &= \max\left(\left[T_{\text{cmax}}^n(\mathbf{x}_1),\ T_{\text{cmax}}^n(\mathbf{x}_2), \ldots, T_{\text{cmax}}^n\left(\mathbf{x}_{N_{par}-1}\right), T_{\text{cmax}}^n\left(\mathbf{x}_{N_{par}}\right)\right]\right), \\ T_{min}^n &= \min\left(\left[T_{\text{cmax}}^n(\mathbf{x}_1),\ T_{\text{cmax}}^n(\mathbf{x}_2), \ldots, T_{\text{cmax}}^n\left(\mathbf{x}_{N_{par}-1}\right), T_{\text{cmax}}^n\left(\mathbf{x}_{N_{par}}\right)\right]\right), \\ T_{mean}^n &= \frac{1}{N_{par}}\sum_{i=1}^{N_{par}} T_{\text{cmax}}^n(\mathbf{x}_i). \end{aligned} \tag{22}$$

The $T_{\mathrm{cmax}}(\mathbf{x}_i^n)$ temperatures are calculated for all particles existing in a swarm, i.e. $i = 1,...,N_{par}$. The temperature distribution in underground power cable systems determined from FEM simulations (subsection 4.2) is used for this purpose.

Then for iteration $n = 1$ until the algorithm convergence is achieved, the following algorithm steps are performed within a single iteration:

1. Determination of the *pbest* (particle best) and *gbest* (global best) particle positions.
2. Calculation of the delta momentum term $\Delta\mathbf{v}_i^{n-1} = \mathbf{v}_i^{n-1} - \mathbf{v}_i^{n-2}$ and next the particles velocities $\mathbf{v}_i^n$ from Eq. (1) by using the scaled design variables vectors $\widehat{\mathbf{x}}_i^{n-1}$ for $i = 1,...,N_{par}$.
3. Updating a scaled design variables vectors $\widehat{\mathbf{x}}_i^{n-1}$ to $\widehat{\mathbf{x}}_i^n$ for all particles in the swarm, i.e. $i = 1,...,N_{par}$, using Eq. (2).

It should be noted that in Eqs. (1)–(2), the design variables vectors $\mathbf{x}_i^{n-1}$ for $i = 1,...N_{par}$ are replaced with scaled equivalents i.e. $\widehat{\mathbf{x}}_i^{n-1}$ for $i = 1,... N_{par}$.

4. Rescaling of the design variables vectors $\widehat{\mathbf{x}}_i^n$ to $\mathbf{x}_i^n$, for $i = 1,...,N_{par}$, using Eq. (21).
5. Creation of the underground power systems FEM models based on the values of the rescaled design variables vectors $\mathbf{x}_i^n = [p_i^n, b_i^n, s_i^n, l_i^n]$ for $i = 1,...,N_{par}$.
6. Solving the steady state heat conduction problem, Eq. (10), and determining the$T_{\mathrm{cmax}}(\mathbf{x}_i^n)$ temperatures for $i = 1,...,N_{par}$.
7. Evaluation of the cost function $F(\mathbf{x}_i^n)$, given by Eq. (7) with constraints, Eqs. (8)–(9), for all particles existing in the swarm $i = 1,...,N_{par}$.
8. Calculation of the T_{max}^n, T_{mean}^n and T_{min}^n temperatures from Eq. (22) based on the determined $T_{\mathrm{cmax}}(\mathbf{x}_i^n)$ temperatures.
9. Checking if the convergence criterion $\in < 0.0001$ is satisfied:

$$\boldsymbol{\delta}^n = \left[\delta_1^n, \delta_2^n, ..., \delta_{N_{par}-1}^n, \delta_{N_{par}}^n\right] = \left[\left|\mathbf{v}_1^n\right|, \left|\mathbf{v}_2^n\right|, ..., \left|\mathbf{v}_{N_{par}-1}^n\right|, \left|\mathbf{v}_{N_{par}}^n\right|\right], \quad \in = \max(\boldsymbol{\delta}^n) < 0.0001. \tag{23}$$

In Eq. (23) δ_i^n is equal to the magnitude of the $\mathbf{v}_i^n$ vector. The values of δ_i^n are determined for all particles existing in a swarm, i.e. $i = 1,...,N_{par}$. The maximum value of $\boldsymbol{\delta}^n$ components, i.e. $\in$, shall not exceed 0.0001.

In order to analyze the momentum-type PSO algorithm behavior, 200 iterations are considered when presenting the computational results. This number of momentum-type PSO algorithm iteration allowed to obtain the convergent solution. The swarm sizes are assumed as $N_{par} = 30$ and $N_{par} = 10$. Hence, the influence of swarm size on both the obtained global minimum of $F(\mathbf{x})$ function and the algorithm behavior can be studied.

The cost function $F(\mathbf{x})$ variations with consecutive iterations are shown in Fig. 6a for $N_{par} = 30$ and in Fig. 6c for $N_{par} = 10$. It can be observed that in the final iteration the mean value of $F(\mathbf{x})$, i.e. $F(\mathbf{x})_{mean}$ is lower for $N_{par} = 10$, i.e. $F(\mathbf{x})_{mean} = 11.8$ than $N_{par} = 30$, i.e $F(\mathbf{x})_{mean} = 42.6$. Moreover, the maximum values of $F(\mathbf{x})$, i.e. $F(\mathbf{x})_{max}$, are lower for $N_{par} = 10$ than $N_{par} = 30$. The resultant global minimum of $F(\mathbf{x})$, i.e. the optimum A_b, that corresponds to the $F(\mathbf{x})_{min}$ in a final iteration, is equal to 0.3196 for $N_{par} = 10$ and 0.3195 for $N_{par} = 30$. Therefore, the similar results are obtained for the lower number of particles i.e. $N_{par} = 10$, than for the larger i.e. $N_{par} = 30$ (Figs. 6a, c, Table 6).

Fig. 6b and Fig. 6d present variations of T_{max}, T_{mim} and T_{mean} values with the increasing iteration number for $N_{par} = 30$ and $N_{par} = 10$, respectively. The fluctuations in T_{max} or T_{min}, which may be observed in Fig. 6b and Fig. 6d, indicate that the algorithm works correctly, avoiding the local minimum of the modified objective function $F(\mathbf{x})$. The dynamic penalty function $F_{pen} = C(n)H(\mathbf{x})$ (Eq. (7)) acts stronger with successive iterations, ensuring that the obtained conductor temperatures T_{cmax} approach to the optimum

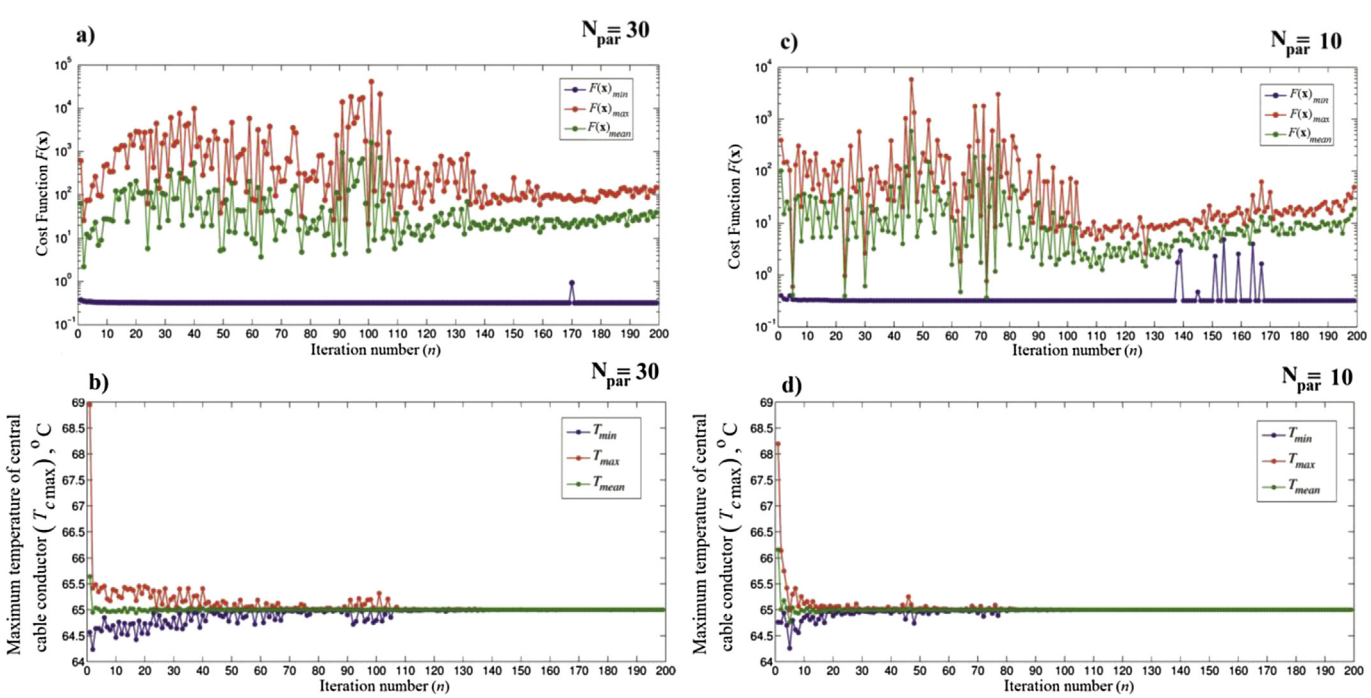

Fig. 6. Variation of maximum (*max*), arithmetic average (*mean*) and minimum (*min*) values of $F(\mathbf{x})$ and T_c with consecutive iteration number for $N_{par} = 30$ (Figs. 6a, b) and $N_{par} = 10$ (Figs. 6c, d).

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Table 6
Optimum values of design variables l, s, b, and p (Fig. 3), maximum conductor temperature $T_{cmax} = T_{mean}^{n=200}$, and FTB bedding layer cross-sectional area A_b.

N_{par}	l, m	s, m	b, m	p, m	T_{cmax}, °C	A_b, m^2
30	0.6	0.2	0.31327	0.20044	65	0.3195
10	0.6	0.2	0.31366	0.20000	65	0.3196

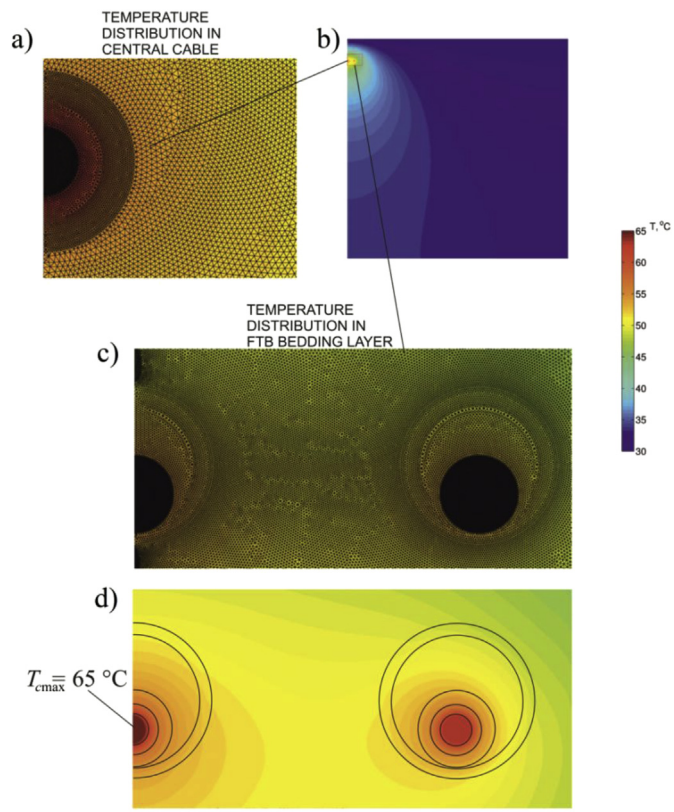

Fig. 7. Temperature distribution in: a) vicinity of the central cable, b) analyzed underground power cable system (cable, bedding, mother ground), (c–d) FTB bedding, HDPE pipes, SBM mixture and cable.

value $T_{opt} = 65$ °C for the entire particle swarm. The global best values of the design variables, i.e. l, s, b, and p, the corresponding area of FTB layer A_b and the obtained maximum conductor temperature T_{cmax} are listed in Table 6.

From Table 6 one can deduce, that the particles number $N_{par} = 10$ is sufficient to explore the solution domain. The increase in a number of particles to $N_{par} = 30$ does not allow to obtain considerably better solution. It should be noted that the increase in swarm size N_{par} results in a significant computational overhead. The computational time for $N_{par} = 30$ and 200 iterations is over three times longer than for $N_{par} = 10$.

Fig. 7 presents the temperature distribution obtained for the optimum values of design variables. The maximum temperature $T_{cmax} = 65$ °C occurs in the central cable core (Fig. 7a). Fig. 7b depicts the temperature distribution in the entire underground power cable system (Fig. 3). Fig. 7c shows the FEM mesh for FTB layer, HDPE casing pipes and power cables used in the computations, when optimal shape of FTB bedding layer is achieved. The temperature distribution obtained for FTB bedding layer, HDPE casing pipes, and power cables are depicted in Fig. 7d.

The heat is generated in the central and side cable conductors. Due to the cable temperature fields interactions and worse heat dissipation conditions, the central cable conductor temperature is higher than the temperature of side cable conductors.

6. Summary

This study presents the application of momentum-type Particle Swarm Optimization (PSO) for the thermal performance optimization of underground power cable system. An electrical energy transmission line that consists of three underground power cables, placed in a flat (in-line) arrangement, is considered. To avoid the excessive mechanical loads the cables are laid in HDPE casing pipes, filled with SBM. The HDPE casing pipes are buried directly in Fluidized Thermal Backfill (FTB) layer to protect the cables from overheating. Due to the substantial costs of FTB backfill material (about $130 per cubic meter), the cross-sectional area of FTB bedding layer need to be minimized. The momentum-type Particle Swarm Optimization (PSO) algorithm is used for this purpose.

The geometrical dimensions of FTB bedding layer and the cable spacing are used as design variables in the optimization procedure. The particle swarms that consists of $N_{par} = 10$ and $N_{par} = 30$ particles, containing the values of design variables, are considered in the computations. The behavior of the following design variables is studied during the algorithm operation (Fig. 4b):

- cable spacing l,
- distance from the cable to the top edge of FTB bedding layer b,
- distance from the cable to the bottom edge of FTB bedding layer p,
- distance from the cable to the right edge of FTB bedding layer s.

The results comparison for the different swarm sizes allowed to check its influence on the obtained optimum solution. The iteration number considered in the PSO optimization is 200. The temperature distribution in the cable system is obtained by means of the Finite Element Method under the assumption that resulting temperature field is steady-state and two-dimensional. Determined maximum temperature of the cable conductor is compared with the optimal temperature for cable operation, given by the cable producer, (i.e. 65 °C). The computations considered the current loading and volumetric heat generation rate from the cable conductor as temperature-dependent. The cable conductor maximum temperature determined from FEM model shall not exceed the optimum temperature of cable operation. Therefore, PSO algorithm employs the dynamic penalty function which eliminates the particles at a higher temperature than the optimal one from the swarm.

Although the patterns of design variables variation with increasing iteration number are different the obtained final design variables values are practically the same for $N_{par} = 10$ and $N_{par} = 30$:

- cable spacing l equal to 0.600 m;
- height and width of the FTB bedding cross-section are equal to 0.5133 m and 1.6 m, respectively;
- maximum cable core temperature, according to producer recommendation, $T_{opt} = 65$ °C.

The physical geometry of the modeled underground power cable system and the expected pattern of the thermal solution reveal the mirror symmetry. Therefore, $A_b = 0.3195$ m^2 obtained from the optimization is, in fact, only a half of the real FTB cross-sectional area, which is equal to 0.6390 m^2. It should be noted that the exact value of width for FTB bedding layer is two times larger than obtained from the optimization since the symmetry boundary condition is applied in the FEM model.

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In view of the results obtained in this study, a momentum-type Particle Swarm Optimization may be regarded as an efficient tool for optimizing the underground power cable systems.

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